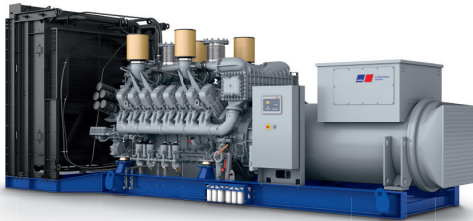




Diesel Generator Set

mtu 16V4000 DS2250

380V – 11 kV/50 Hz/standby power/NEA (ORDE) optimized
16V4000G74F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2230 kVA - 2350 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)		
Manufacturer	mtu		Total oil system capacity: l	300	
Model	16V4000G74F		Engine jacket water capacity: l	175	
Type	4-cycle		Intercooler coolant capacity: l	50	
Arrangement	16V		Combustion air requirements		
Displacement: l	76.3				
Bore: mm	170			Combustion air volume: m³/s	2.5
Stroke: mm	210			Max. air intake restriction: mbar	50
Compression ratio	16.4		Cooling/radiator system		
Rated speed: rpm	1500				
Engine governor	ECU 9			Coolant flow rate (HT circuit): m³/hr	68.5
Max power: kWm	1965			Coolant flow rate (LT circuit): m³/hr	30
Air cleaner	dry		Heat rejection to coolant: kW	660	
Fuel system			Heat radiated to charge air cooling: kW	430	
			Heat radiated to ambient: kW	90	
			Fan power for electr. radiator (40°C): kW	70	
Maximum fuel lift: m	5		Exhaust system		
Total fuel flow: l/min	20				
Fuel consumption ²⁾	l/hr g/kwh			Exhaust gas temp. (after turbocharger): °C	480
				Exhaust gas volume: m³/s	6.6
			Maximum allowable back pressure: mbar	85	
			Minimum allowable back pressure: mbar	30	
At 100% of power rating:	471.1	199			
At 75% of power rating:	358.7	202			
At 50% of power rating:	247.4	209			

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1864	2330	3540	1808	2260	3434
	400 V	1864	2330	3363	1808	2260	3262
	415 V	1864	2330	3242	1808	2260	3144
Leroy Somer LSA52.3 L12 (Low voltage Leroy Somer oversized)	380 V	1864	2330	3540	1808	2260	3434
	400 V	1864	2330	3363	1808	2260	3262
	415 V	1864	2330	3242	1808	2260	3144
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1848	2310	3510	1784	2230	3388
	400 V	1848	2310	3334	1784	2230	3219
	415 V	1808	2260	3144	1784	2230	3102
Marathon 1020FDL7093 (Low voltage Marathon oversized)	380 V	1848	2310	3510	1784	2230	3388
	400 V	1848	2310	3334	1784	2230	3219
	415 V	1808	2260	3144	1784	2230	3102
Marathon 1020FDH7097 (Medium volt. marathon)	11 kV	1864	2330	122	1800	2250	118
Leroy Somer LSA53.2 XL9 (Medium volt. Leroy Somer)	11 kV	1880	2350	123	1808	2260	119

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.